

# Customer Segmentation for Malls Using Machine Learning

# Business Problem and Objective

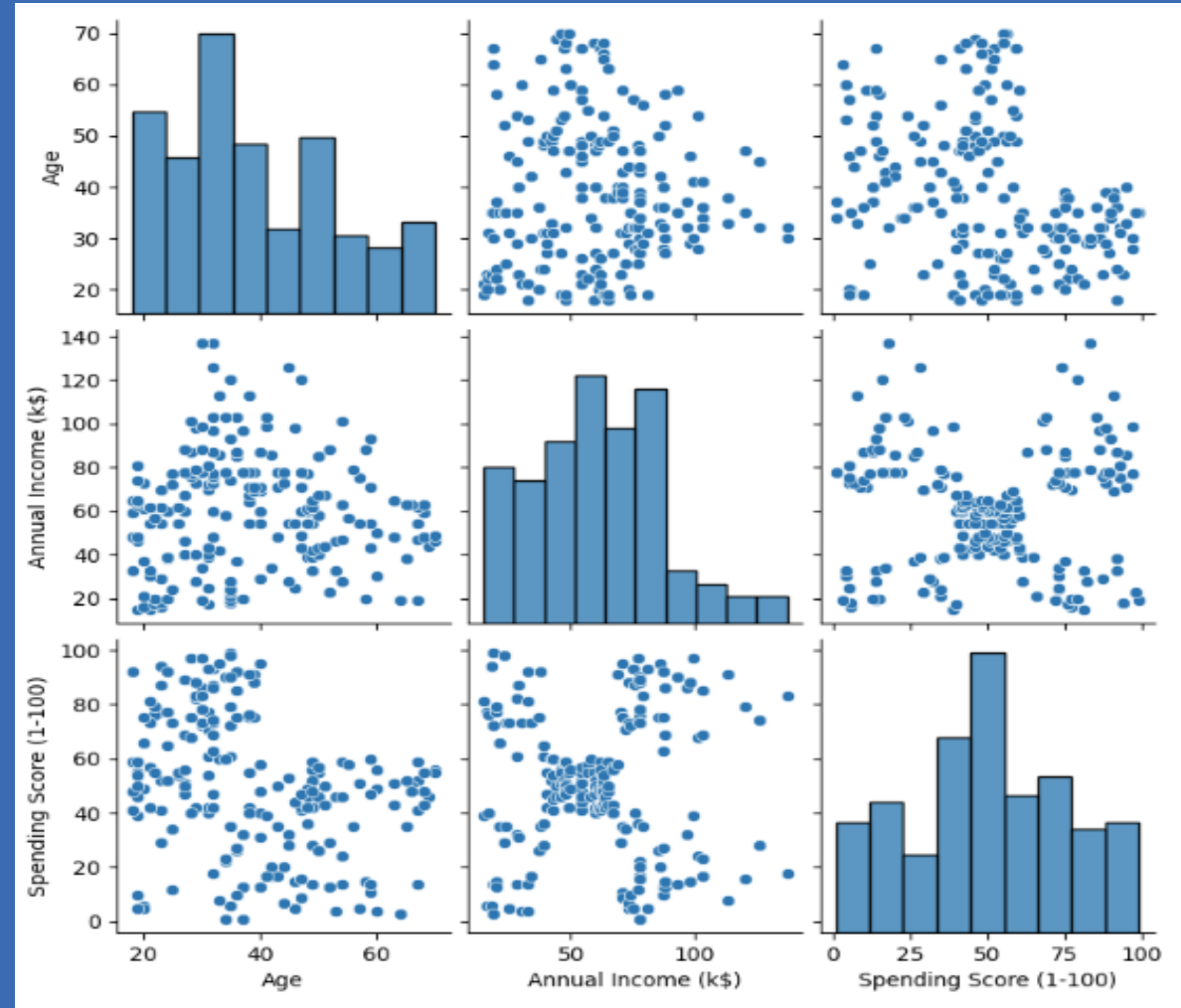
- Diverse mall customers with varying spending and income
- Traditional segmentation limited to demographics
- Objective: Identify meaningful customer segments using data
- Goal: Enable personalized marketing and improve engagement

# Dataset and Preprocessing

- Source: Kaggle – Mall Customer Segmentation Data
- Real-world mall customer records
- Features: Age, Gender, Income, Spending Score
- Gender excluded to avoid bias
- Dropped CustomerID (non-predictive)
- No missing values
- Outliers removed using Interquartile Range (IQR) method

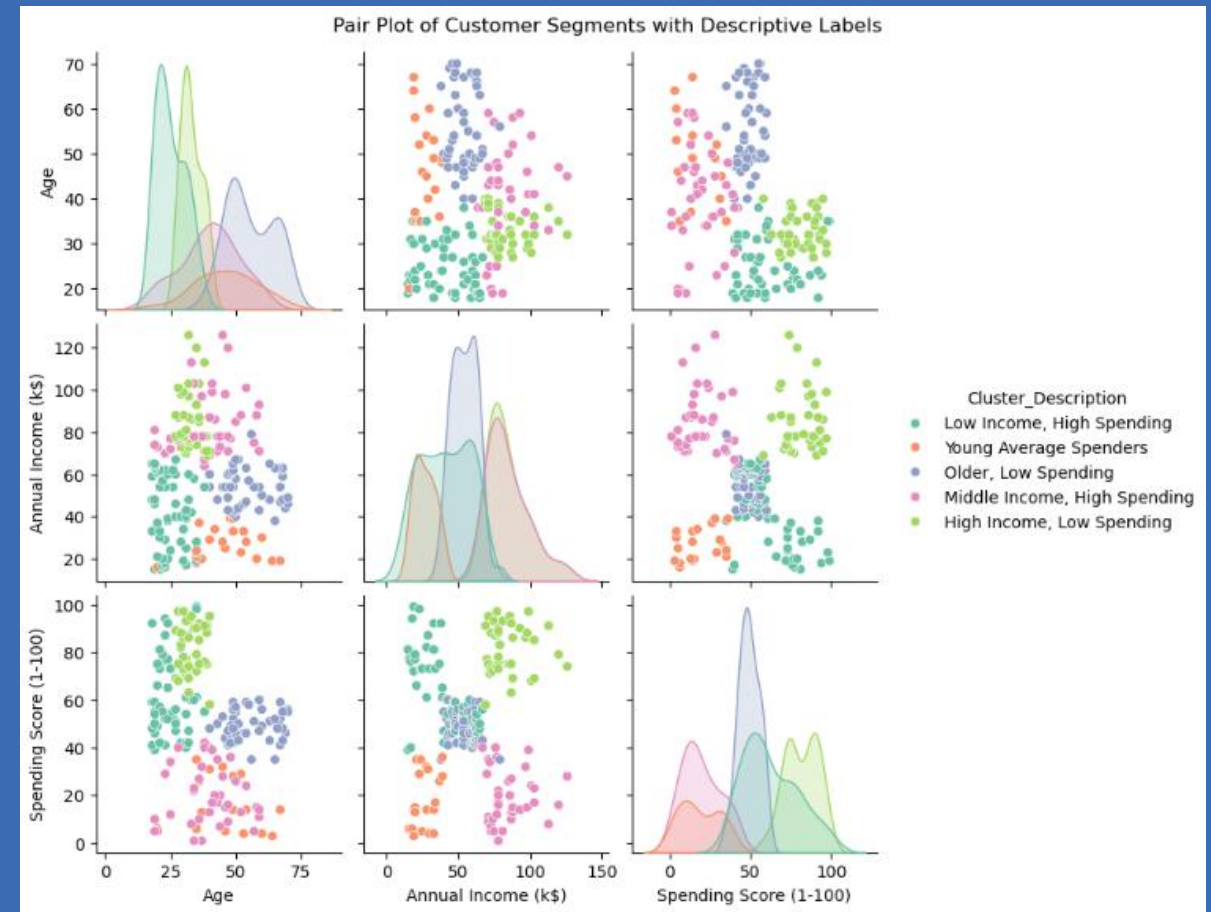
# Exploratory Data Analysis (EDA)

- Most customers are young adults (20s–30s)
- Income centered around middle-income levels
- Spending Score shows clear high/low groupings
- Pairplot shows natural clusters based on behavior



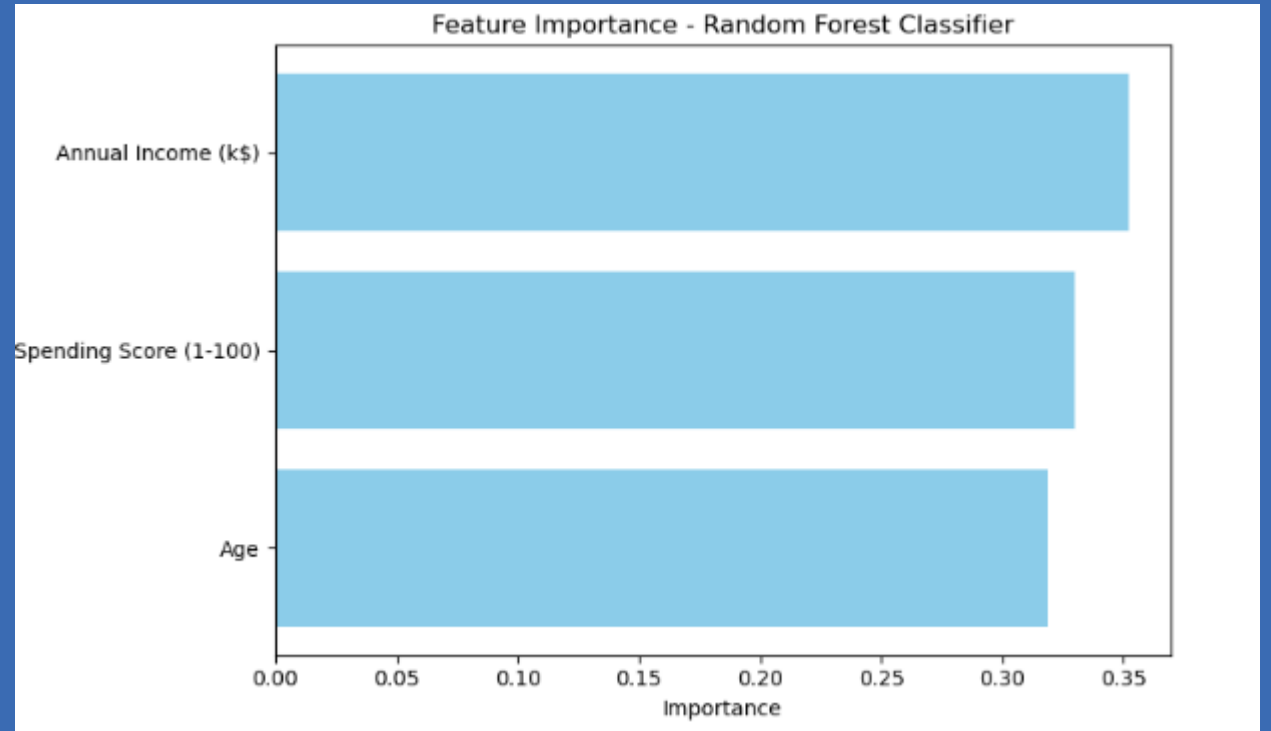
# Unsupervised Learning – KMeans Clustering

- Standardized features to equalize influence
- Used Elbow Method to select 5 clusters
- Identified distinct groups:
  - *Low Income, High Spending*
  - *High Income, Low Spending*
  - *Young Average Spenders*
  - *Middle Income, High Spending*
  - *Older, Low Spending*



# Supervised Learning – Random Forest Classification

- Used cluster labels as target classes
- Trained Random Forest to predict segments for new customers
- Achieved perfect accuracy on test data (cautiously interpreted)
- Feature importance: Income and Spending Score were key drivers



# Implementation and Use Cases

- Integrate segmentation model into mall CRM systems
- Routine data preprocessing and automated segment assignment
- Enable marketing teams to craft personalized campaigns
- Monitor segments over time to adapt strategies
- Potential to incorporate more data features (purchase history, online behavior)

# Limitations and Ethical Concerns

- Small dataset may limit broad conclusions
- Spending Score is subjective, not exact spending
- Perfect classification may reflect overfitting
- Missing features like visit frequency or purchases
- Gender excluded to avoid bias
- No personal identifiers used



# Conclusion & Final Thoughts

- Effective segmentation reveals actionable customer groups
- Clustering combined with classification enables scalable marketing insights
- Financial and behavioral features drive segmentation
- Approach can scale with more data and features for real-world application

# References

- vjchoudhary7. (n.d.). Customer Segmentation Tutorial in Python. Kaggle. <https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>
- scikit-learn: machine learning in Python — scikit-learn 1.7.0 documentation. (n.d.). <https://scikit-learn.org/stable/>
- Jain, A. K. (2009). Data clustering: 50 years beyond K-means. Pattern Recognition Letters, 31(8), 651–666. <https://doi.org/10.1016/j.patrec.2009.09.011>